

Unload a Dish Rack into a Cabinet Organizer SOP

One episode empties one loaded dish rack into one cabinet organizer. Nine washed items sit in the rack, still damp from the wash: a large plate, a small plate, two bowls, two cups with handles, a fork, a knife, and a spoon. Every one of them is dried on a towel before it is put away, and every one of them has a fixed place in the organizer.

The rack is loaded differently every episode. Which slot each plate stands in, where the bowls and cups sit, and whether something is resting over something else all change from take to take. So **the unload order is fixed by item type, not by position**: plates, then bowls, then cups, then cutlery. That one rule is what keeps the data consistent across a scene that is never the same twice.

Every item travels the same route: out of the rack, onto the drying mat in the middle of the table, wiped there with the towel, then into its place in the organizer on the left. No item skips the mat and no item skips the wipe.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the cabinet organizer on the left, the drying mat in the middle, the towel in its home spot, and the loaded dish rack on the right.
3. Both arms are at home with grippers open.
4. The tabletop is dry outside the mat and the rack.
5. Nothing obstructs the paths between the rack, the drying mat, and the organizer.

Materials checklist

1. The dish rack sits against the right edge, loaded with all nine items: a large plate, a small plate, two bowls, two cups with handles, a fork, a knife, and a spoon.
2. Plates stand upright in the rack slots. Bowls and cups sit inverted on the rack shelf. The three utensils stand handle up in the rack basket.
3. Every item is damp. No item is dripping and there is no standing water in the rack tray.
4. The load is uneven, as a real wash leaves it: items at no fixed spacing, and at least one bowl or cup sitting partly over a plate.
5. The wiping towel is dry, folded into a flat pad, and lying in its home spot to the right of the drying mat.

6. The drying mat lies flat in the middle of the tabletop and is empty.
7. The cabinet organizer is empty: nothing in the plate insert, nothing in the bowl insert, both cup hooks bare, and all three cutlery compartments bare.

Workspace layout

- **Cabinet organizer:** on the tabletop against the left edge, within reach of the left arm. It holds the plate insert and the bowl insert side by side, the cutlery tray along its front edge, and the cup rail with two hooks along its back edge.
- **Drying mat:** flat in the middle of the tabletop, reachable by both arms. It holds exactly one item at a time and is empty at the end of the episode.
- **Towel home:** the clear patch of tabletop just right of the drying mat, within reach of the right arm. The towel lies here whenever it is not being used.
- **Dish rack:** on the tabletop against the right edge, within reach of the right arm.

Each arm owns the side it sits on. The right arm works the rack and the towel, the left arm works the organizer, and every item changes arms by resting on the drying mat.

Vocabulary

Arm assignments

- **The right gripper owns the rack and the towel.** It lifts every item out of the rack, sets it on the drying mat, and does every wipe. It never reaches the organizer.
- **The left gripper owns the organizer.** It holds each item still on the mat during the wipe, then carries it into the organizer. It never reaches the rack and never touches the towel.
- **No item is passed hand to hand.** Every item changes arms by being set down on the drying mat and picked up again from there.
- An item is released only onto the drying mat or into its place in the organizer. Never open a gripper over open space, and never over the rack once an item has left it.

The drying cycle

Every one of the nine items goes through the same cycle. The step for each item type gives the grip, the drying pose, and the place it goes, but the shape of the cycle never changes. It has three parts.

Part 1, place it. The right gripper lifts the item straight up out of the rack, carries it level to the drying mat, and lowers it into the drying pose for its type. It releases once the item is resting still.

Part 2, wipe it. The left gripper takes hold of the item on the mat and holds it still. The right gripper pinches the towel in its home spot, lifts it, presses it onto the item, and draws it across

the item **twice**. It then lays the towel back flat in its home spot and releases. The left gripper keeps its hold the whole time.

Two passes is the standard, not a dry surface. Whatever moisture is left after two passes stays.

Part 3, put it away. The left gripper, still holding the item from the wipe, lifts it off the mat, carries it level into the organizer, lowers it into its place, and releases.

Grip each item once. The hold the left gripper takes for the wipe is the hold it carries the item by. Do not let go and take a fresh grip between the wipe and the put away.

Working a loaded rack

- **Type order, not position order.** Unload plates, then bowls, then cups, then cutlery, no matter where each item happens to be sitting. Do not unload by whatever is nearest or easiest.
- Within the plates, the order is always **large plate, then small plate**. Within the bowls, the cups, and the cutlery, take the item **nearer the front of the rack first**. Cutlery is the one exception to front first: it always goes **fork, knife, spoon**.
- **Never pull an item out from under another.** If the item due next is covered or blocked, lift the covering item straight up first, set it clear on the rack shelf, and then take the item that was due. This is the only permitted departure from the order, and the order resumes from the top of the type you were on.
- Take each grip where the item is clear, not where the standard grip would be on an empty rack.
- Never drag an item along a rack slot or across the rack shelf to get a better grip. Lift it straight up.

Organizer places

- **Plates:** stacked flat and face up in the plate insert, large plate on the bottom, small plate centered on top of it.
- **Bowls:** nested face up in the bowl insert, the first bowl down flat and the second bowl lowered inside it.
- **Cups:** hanging rim down from the cup rail, handle over the hook. The first cup goes on the near hook, the second on the far hook.
- **Cutlery:** lying flat in the cutlery tray, one utensil per compartment. The fork goes in the compartment nearest the left edge, the knife in the middle compartment, the spoon in the far compartment. All three handles point toward the front of the tray.

Nothing rests on the organizer outside these places, and nothing overhangs its edge.

Handling standard

- Move exactly one item at a time.

- The towel is never carried in the same gripper as an item, and never in the same trip.
- Carry every item level and lower it straight down.
- Lower each item until it is resting on the mat, on the insert, or on the item below it, then release. Never drop an item from above.
- Nothing is slid. The only sliding motion in the whole episode is the towel drawn across an item during a wipe. Correct a bad placement by lifting the item and setting it down again.
- An item is settled when it stays still for 2 seconds after release. A hung cup is settled when it has stopped swinging.

Steps

Step 1: Dry and stack the two plates

Goal: both plates are dry and stacked face up in the plate insert.

Work one plate at a time in this order, wherever each one is standing: **large plate, then small plate.**

1.1 Place it on the mat

- If a bowl or cup is resting over the plate due next, lift that item straight up and set it clear on the rack shelf first.
- The **right gripper** grips the plate by any free stretch of rim, lifts it straight up out of the slot, and carries it level to the drying mat.
- Lower it flat and face up in the middle of the mat, then release.

1.2 Wipe it

- The **left gripper** takes the plate by its rim, the same hold it will carry it by, and holds it still against the mat.
- The **right gripper** lifts the towel from its home spot, presses it onto the face of the plate, and draws it across the face twice.
- The **right gripper** lays the towel back flat in its home spot and releases.

1.3 Stack it in the plate insert

- The **left gripper** lifts the plate off the mat and carries it level to the organizer.
- Lower it face up into the plate insert. The large plate goes flat on the bottom of the insert; the small plate is centered on top of the large one.
- Release once the plate is resting.

Check: both plates are face up in the insert, large on the bottom, small centered on top, and the stack sits square rather than leaning. If it leans or the top plate rocks, lift that plate and lower it again. Do not shove it into place.

Step 2: Dry and nest the two bowls

Goal: both bowls are dry and nested face up in the bowl insert.

Work one bowl at a time, taking the **bowl nearer the front of the rack first**.

- **Place:** the **right gripper** grips the bowl by its rim, lifts it straight up off the rack shelf, carries it to the drying mat, and lowers it face up in the middle of the mat.
- **Wipe:** the **left gripper** takes the bowl by its rim and holds it still. The **right gripper** draws the towel across the outside of the bowl twice, then returns the towel to its home spot.
- **Put away:** the **left gripper** lifts the bowl, carries it level to the organizer, and lowers it face up into the bowl insert. The first bowl goes flat in the insert; the second is lowered inside the first until it is resting on it.

Check: both bowls are face up in the insert and the second sits down inside the first rather than perched on its rim. If the second bowl sits high or tips, lift it and lower it again.

Expected state: the rack still holds the two cups and the three utensils. The mat is bare, the towel is flat in its home spot, and the plate insert and bowl insert are both loaded.

Step 3: Dry and hang the two cups

Goal: both cups are dry and hanging rim down from the cup rail.

Work one cup at a time, taking the **cup nearer the front of the rack first**. The first cup goes on the near hook, the second on the far hook.

3.1 Place it on the mat

- The **right gripper** grips the cup around its body, lifts it straight up off the rack shelf, and carries it level to the drying mat.
- Lower it **rim down** in the middle of the mat with the **handle pointing left**, toward the organizer, then release.

3.2 Wipe it

- The **left gripper** takes the cup around its body, the same hold it will carry it by, and holds it still.
- The **right gripper** draws the towel across the base and down the side of the cup twice, then lays the towel back in its home spot.

3.3 Hang it

- The **left gripper** lifts the cup off the mat, keeping it rim down and the handle pointing away from the arm, and carries it to the cup rail.
- Lower the cup so the handle drops over the hook, then release only once the cup is hanging and has stopped swinging.

- If the handle misses the hook or the cup swings off, catch it or lift it clear, return to the mat if it has been set down anywhere else, and hang it again.

Check: both cups hang rim down and still, the first on the near hook and the second on the far hook. Neither cup is resting on the organizer shelf.

Step 4: Dry and sort the three utensils

Goal: the fork, knife, and spoon are dry and lying one per compartment in the cutlery tray.

Work one utensil at a time in this order, wherever each one is standing in the basket: **fork, knife, spoon.**

- **Place:** the **right gripper** grips the utensil across its handle, lifts it straight up out of the basket, and lays it flat on the drying mat with the **handle pointing right**, toward the rack.
- **Wipe:** the **left gripper** takes the utensil by its handle and holds it flat on the mat. The **right gripper** draws the towel along the utensil from handle to tip twice, then returns the towel to its home spot.
- **Put away:** the **left gripper** lifts the utensil and lays it flat in its compartment: the fork in the compartment nearest the left edge, the knife in the middle, the spoon in the far compartment. All three handles point toward the front of the tray.

Check: one utensil per compartment, in the prescribed compartments, all lying flat with their handles pointing the same way. No compartment holds two utensils.

Expected state: the rack is empty, the mat is bare, the towel is flat in its home spot, and the organizer holds the plate stack, the nested bowls, the two hanging cups, and the three sorted utensils.

Step 5: End the episode

Goal: the rack is empty, the organizer is fully loaded, and both arms are safely home.

- Confirm the rack holds nothing, including the basket, and that no item was left on the mat or on the bare tabletop.
- Confirm the towel is lying flat in its home spot and not on the mat, in the rack, or in the organizer.
- Confirm every place in the organizer is filled as prescribed.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Unload the organizer completely: plate insert, bowl insert, both hooks, all three cutlery compartments.

2. Wet all nine items again so they are damp but not dripping, and shake off any standing water.
3. Reload the rack: plates upright in the slots, bowls and cups inverted on the shelf, the three utensils handle up in the basket. Vary the arrangement between episodes rather than rebuilding the same load, and leave at least one bowl or cup sitting partly over a plate.
4. Empty the rack tray if water has collected in it.
5. Replace the wiping towel with a dry one, folded into a flat pad, and lay it in its home spot.
6. Wipe the tabletop and clear the drying mat.
7. Clean or replace anything chipped, cracked, bent, or stained.
8. Put the organizer back against the left edge and the rack back against the right edge.
9. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Unloaded by position instead of by type

- **Visible cue:** items come out of the rack in whatever order they happen to be reachable, such as a cup before the plates or the cutlery split around a bowl, rather than plates, then bowls, then cups, then cutlery.
- **SOP rule broken:** Steps 1 to 4, unload by type order: plates, bowls, cups, cutlery, no matter where each item is sitting.
- **Coaching note:** the load changes every episode; the order never does.

Violation: Wrong order within a type

- **Visible cue:** the small plate is taken before the large one, the bowl or cup at the back of the rack is taken before the one at the front, or the utensils leave in any order other than fork, knife, spoon.

- **SOP rule broken:** Steps 1 to 4, large plate then small plate, bowls and cups front of the rack first, and cutlery always fork, knife, spoon.
- **Coaching note:** name the next item before you reach for it.

Violation: Item pulled out from under another

- **Visible cue:** a plate is pulled out from under a bowl or cup resting over it, or an item is slid sideways along a slot or across the rack shelf to free it, instead of the covering item being lifted straight up and set clear first.
- **SOP rule broken:** Step 1.1, never pull an item out from under another, lift the covering item straight up, set it clear on the rack shelf, then take the item that was due.
- **Coaching note:** uncover it, then lift it. Never tug it free.

Violation: Arm crossed into the other arm's side

- **Visible cue:** the left gripper reaches into the rack, lifts an item out of it, or picks up the towel; or the right gripper reaches the organizer or puts an item away.
- **SOP rule broken:** Steps 1 to 4, the right gripper owns the rack and the towel, the left gripper owns the organizer, and items change arms only on the drying mat.
- **Coaching note:** right arm right of the mat, left arm left of it.

Violation: Item passed hand to hand

- **Visible cue:** an item is transferred directly from one gripper to the other in mid air instead of being set down on the drying mat by the right arm and picked up from there by the left.
- **SOP rule broken:** Steps 1 to 4, no item is passed hand to hand, every item changes arms by resting on the drying mat.
- **Coaching note:** the mat is the handoff. Nothing changes hands in the air.

Violation: Item put away without being dried

- **Visible cue:** an item goes from the rack into the organizer without reaching the drying mat, or it reaches the mat but is picked up and put away with no towel pass at all.
- **SOP rule broken:** Steps 1 to 4, every item is placed on the mat, wiped there, and only then put away.
- **Coaching note:** no item skips the mat and no item skips the towel.

Violation: Wrong number of towel passes

- **Visible cue:** the towel is drawn across the item once, or three or more times, instead of exactly twice; or the wiping continues to chase a dry surface.
- **SOP rule broken:** Steps 1 to 4, draw the towel across the item twice, two passes is the standard, not a dry surface.
- **Coaching note:** count the two passes, then stop, whatever the item still looks like.

Violation: Item not held during the wipe

- **Visible cue:** the right gripper wipes an item that no gripper is holding, and the item shifts, spins, or is pushed across the mat by the towel.
- **SOP rule broken:** Steps 1 to 4, the left gripper takes hold of the item, holds it still, and keeps its hold for the whole wipe.
- **Coaching note:** the left hand anchors the item before the towel touches it.

Violation: Item regripped between the wipe and the put away

- **Visible cue:** the left gripper releases the item after the wipe and takes a fresh grip before carrying it, or hands it back to the right arm to be repositioned.
- **SOP rule broken:** Steps 1 to 4, grip each item once, the hold taken for the wipe is the hold it is carried by.
- **Coaching note:** take the carrying grip first, then wipe, then go.

Violation: Towel mishandled

- **Visible cue:** the towel is left on the drying mat, dropped on the tabletop or floor, carried in the same gripper as an item, put into the rack or the organizer, or left anywhere other than flat in its home spot between wipes.
- **SOP rule broken:** Steps 1 to 4, lay the towel back flat in its home spot after every wipe, and never carry it in the same gripper as an item.
- **Coaching note:** the towel goes home after every single wipe.

Violation: Item put in the wrong place

- **Visible cue:** a plate goes in the bowl insert, a bowl in the plate insert, a cup set into an insert instead of hung, a utensil in the wrong compartment or two utensils sharing one, or any item left on the organizer shelf outside its place.
- **SOP rule broken:** Steps 1 to 4, plates in the plate insert, bowls nested in the bowl insert, cups hung on the rail, and one utensil per prescribed compartment.
- **Coaching note:** every item has one place; say it before you carry it.

Violation: Plate stack built incorrectly

- **Visible cue:** the small plate is on the bottom, a plate is face down, the top plate is clearly off center, or the stack leans and is left that way.
- **SOP rule broken:** Step 1.3 and Organizer places (large plate flat on the bottom, small plate centered face up on top; lift and lower again if it leans or rocks).
- **Coaching note:** square the small plate on the large one before releasing.

Violation: Bowls not nested

- **Visible cue:** the second bowl is set beside the first, perched on its rim, or placed face down, instead of being lowered inside it.
- **SOP rule broken:** Step 2 and Organizer places (bowls nested face up, the second lowered inside the first until it rests on it).
- **Coaching note:** take the second bowl all the way down inside the first.

Violation: Cup hung incorrectly

- **Visible cue:** a cup is left standing on the organizer shelf, hung rim up, hung on the wrong hook for its order, or released while it is still swinging or before the handle is over the hook.
- **SOP rule broken:** Step 3.3 and Organizer places (hang rim down with the handle over the hook, first cup on the near hook and second on the far hook, release only once it hangs still).
- **Coaching note:** watch the handle seat on the hook, then wait for it to stop moving.

Violation: Cutlery handles pointing different ways

- **Visible cue:** the three utensils lie in the tray with their handles pointing in different directions, or a utensil lies across two compartments or on top of the tray divider.
- **SOP rule broken:** Step 4 and Organizer places (each utensil lies flat inside its own compartment with all three handles pointing toward the front of the tray).
- **Coaching note:** all three handles face you, every time.

Violation: More than one item moved at once

- **Visible cue:** two utensils, two bowls, or a plate with a cup still resting on it are lifted together, or an item is carried while the towel is still in the other gripper mid wipe.
- **SOP rule broken:** Steps 1 to 4, move exactly one item at a time.
- **Coaching note:** one item per trip, every trip.

Violation: Item dropped instead of lowered

- **Visible cue:** a gripper opens above the mat, an insert, or the tray and the item falls the rest of the way, instead of being lowered until it is resting.
- **SOP rule broken:** Steps 1 to 4, lower each item until it is resting, then release, and never drop an item from above.
- **Coaching note:** take it all the way down before you open the gripper.

Violation: Item slid instead of lifted

- **Visible cue:** a gripper pushes, nudges, or drags an item along a rack slot, across the rack shelf, across the mat, or across the organizer, either to get a better grip or to correct a placement, instead of lifting it clear and lowering it again.

- **SOP rule broken:** Steps 1 to 4, nothing is slid except the towel during a wipe, and correct a bad placement by lifting the item and setting it down again.
- **Coaching note:** lift it clear and put it down again; nothing gets shoved into position.

Violation: Item dropped, collided, or knocked something over

- **Visible cue:** an item falls, strikes the rack, the organizer, or another item on the way, knocks a hanging cup off its hook, topples the plate stack, or comes to rest overhanging the organizer edge.
- **SOP rule broken:** Steps 1 to 4, keep each item controlled on a clear travel path and release only after a stable placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Mat or rack left occupied

- **Visible cue:** the episode reaches Step 5 with an item still on the drying mat, still in the rack or its basket, or sitting on the bare tabletop.
- **SOP rule broken:** Step 5 (the rack holds nothing and the mat is bare before the episode ends).
- **Coaching note:** sweep the rack, the basket, and the mat with your eyes before homing.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a place in the organizer unfilled, the towel out of its home spot, an arm away from home.
- **SOP rule broken:** Step 5 (confirm the empty rack, the bare mat, the towel at home, and the fully loaded organizer; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective object:** a chipped, cracked, or bent item found mid episode, a cup hook that pulls loose, or a rack or organizer that shifts off its zone. Replace it before the next episode.
- **Towel saturated before the episode ends:** the towel stops picking up water even though the prescribed two passes were run. Swap in a dry towel before the next episode.

Annotation subtasks (from SOP)

1. Lift a covering item off a plate and set it clear on the rack shelf
2. Move one plate from the rack to the drying mat
3. Move one bowl from the rack to the drying mat
4. Move one cup from the rack to the drying mat
5. Move one utensil from the rack basket to the drying mat
6. Hold an item still on the mat during the wipe
7. Wipe one item with the towel and return the towel to its home spot
8. Stack one plate in the plate insert
9. Nest one bowl in the bowl insert
10. Hang one cup on the cup rail
11. Lay one utensil in its cutlery compartment
12. Return both arms home and end the episode